How Rich Are Mixture Attitudes?

Theory and Experimental Evidence

En Hua Hu Séverine Toussaert

Nuffield internal seminar – 3 November 2026

Mixtures in economics

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Consumption
bundles of goods

Risk / uncertainty
outcome distributions

Information
Blackwell structures

Time
consumption streams

Social
allocations

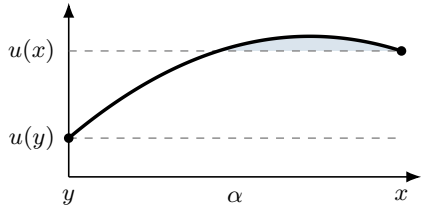
Strategy
mixed strategies

Mixtures in economics

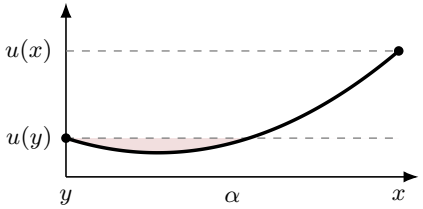
Consumption bundles of goods	**Risk / uncertainty** outcome distributions	**Information** Blackwell structures
Time consumption streams	**Social** allocations	**Strategy** mixed strategies

Most economic environments combine objects through a *mixture operation*

Mixture operation describes preferences

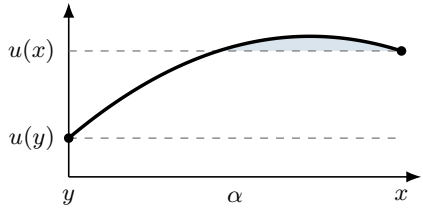


Mixture loving

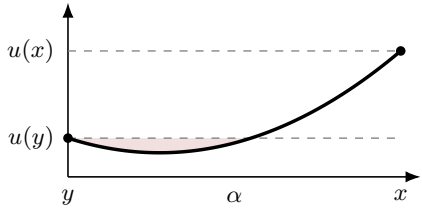


Mixture averse

Mixture operation describes preferences



Mixture loving



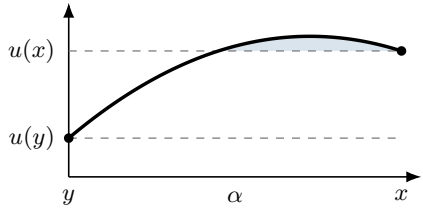
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Most theories assume one-sided attitudes

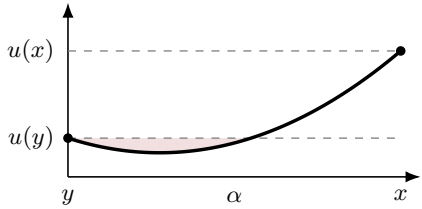
Cobb–Douglas, CES
diversification, smoothing
fairness, maxmin EU

specialization
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Mixture loving



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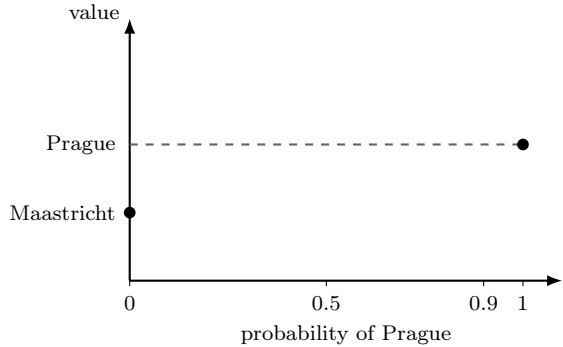
specialization
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This paper: what about preferences that are neither?

Our interest: richer mixture attitudes

Example: surprise trips

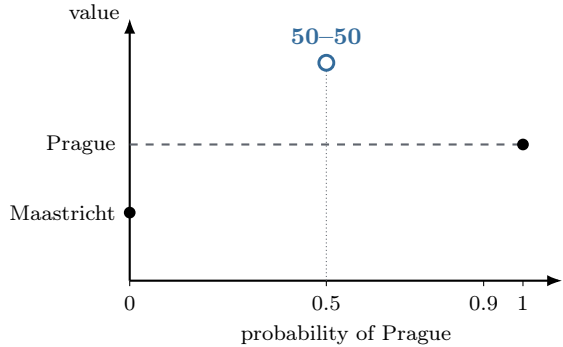
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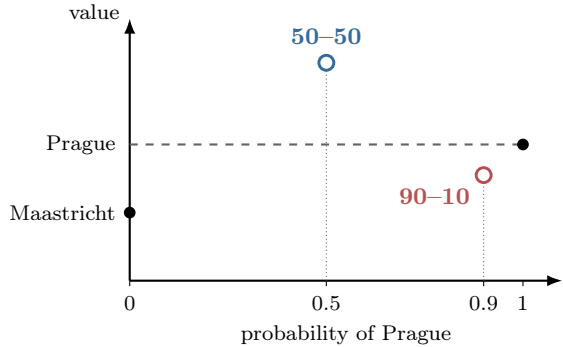
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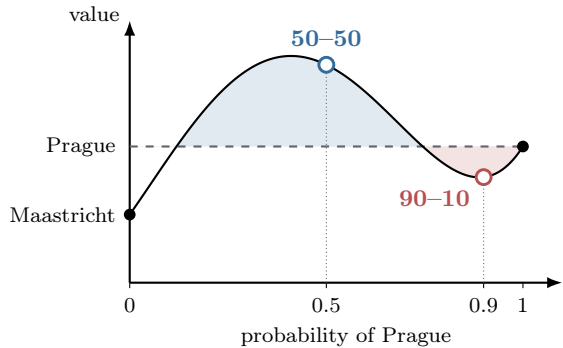
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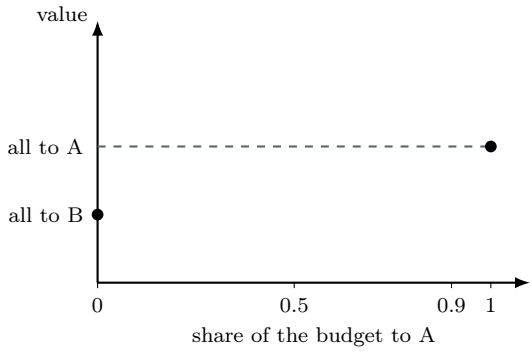


Like to randomize if sufficiently random, else prefer certainty

More examples: giving to charity

Social preferences

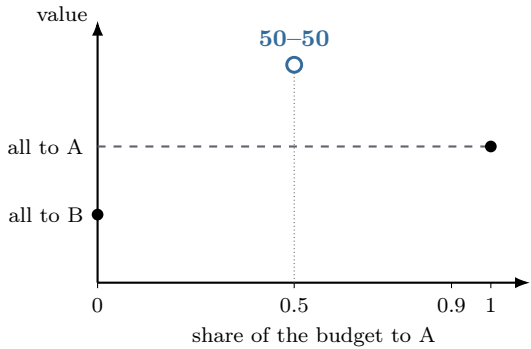
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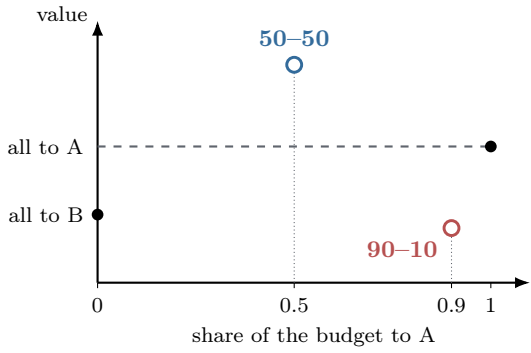
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  fair across causes



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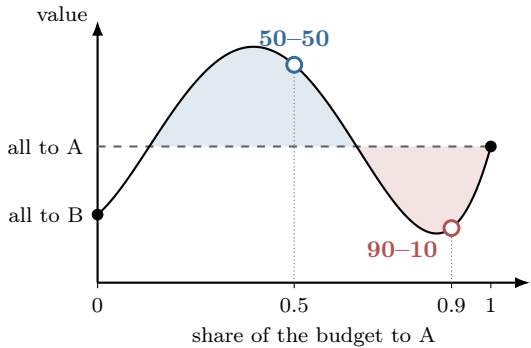
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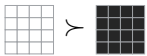
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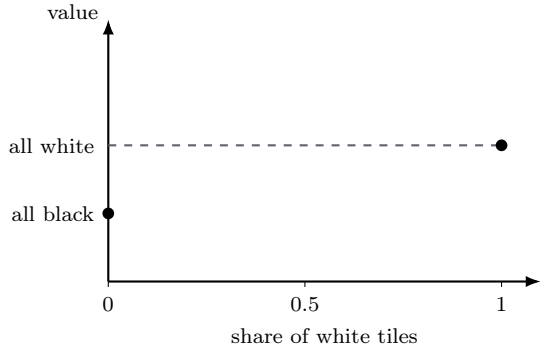


Trade-off between fairness and effectiveness: a small gift to B helps neither

More examples: colour of tiles

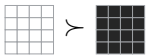
Design

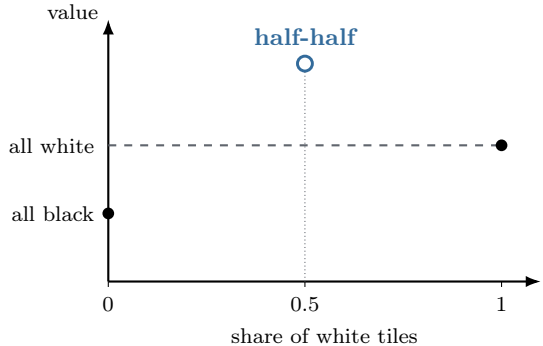
-  white over black



More examples: colour of tiles

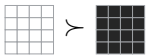
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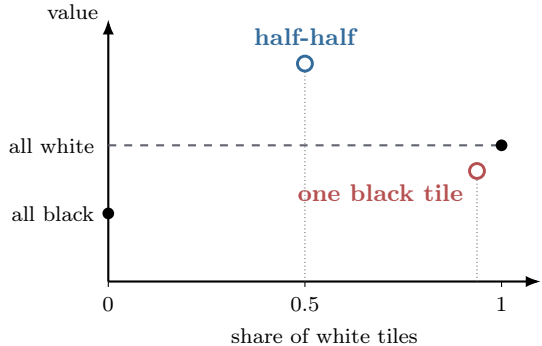
-  white over black
-  a nice half-half pattern



More examples: colour of tiles

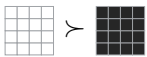
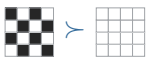
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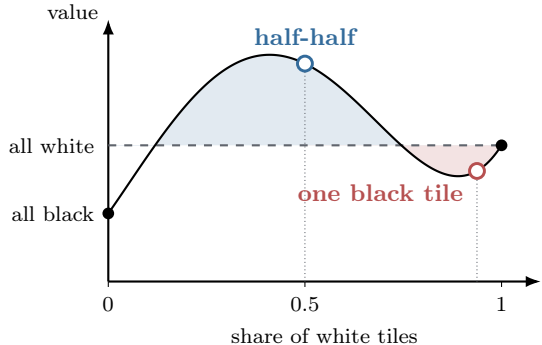
-  white over black
-  a nice half-half pattern
-  a single black tile stands out



More examples: colour of tiles

Design

-  white over black
-  a nice half-half pattern
-  a single black tile stands out



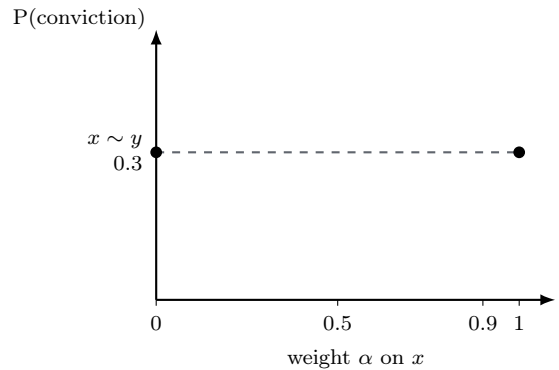
Like a balanced pattern, but a single odd tile spoils the floor

More examples: Bayesian persuasion

Information

Prosecutor and judge (Kamenica & Gentzkow 2011): guilty w.p. 0.3; judge convicts if posterior ≥ 0.5

- $x \sim y$: fully revealing
  labels swapped; conviction w.p. 0.3

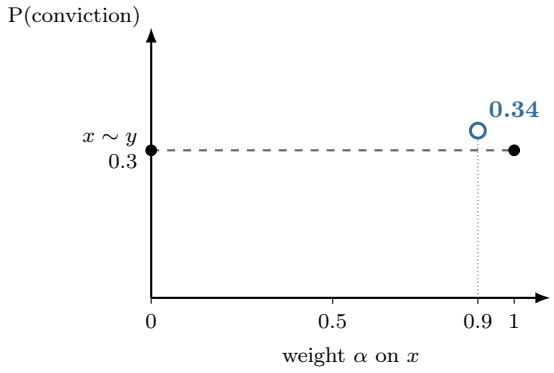


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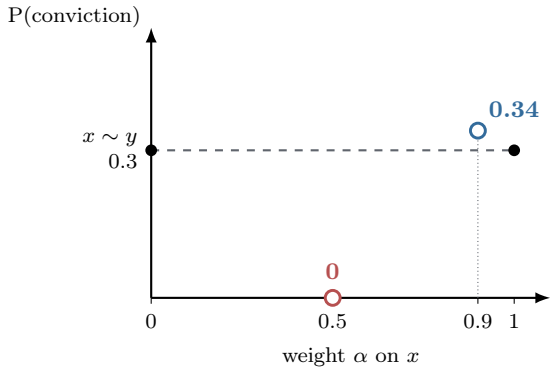


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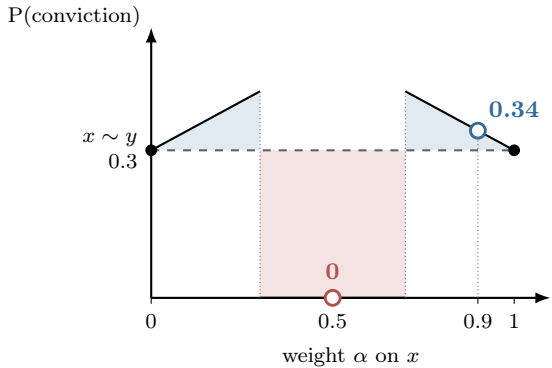


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A little garbling raises conviction; too much leaves the judge uninformed

Why richer mixture attitudes?

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Theory: No conceptual/organizing framework for mixed attitudes

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Theory: No conceptual/organizing framework for mixed attitudes

We build a hierarchy to study and classify mixed attitudes, revealing new insights

Empirics: Limited direct evidence of the prevalence of mixed attitudes

Show mixed attitudes are common and missed in a previously well-studied setting

Related literature and contribution

1. Mixture and convex spaces

Herstein & Milnor (1953); Debreu (1959); Gilboa & Schmeidler (1989); Fehr & Schmidt (1999)

Contribution: hierarchy of mixture attitudes for general spaces. Literature audit.

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3. Preferences for randomization and for certainty

Allais (1953); Loewenstein (1987); Agranov & Ortoleva (2017); Cerreia-Vioglio et al. (2019)

Contribution: documenting joint behavior. General insight: mechanisms that generate mixed attitudes.

Theory

Framework

- **Objects:** compact convex $\mathcal{X} \subseteq [0,1]^d$ containing the diagonal $(t, \ldots, t)$
- **Mixing operation:** $x \alpha y = \alpha x + (1 - \alpha) y$

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Environment	**Object** x	**Mixture** $x \alpha y$	**Diagonal** $(t, \ldots, t)$
Bundles	quantities of d goods	average each good	t of every good
Allocations	payoffs of d people	average each payoff	equal split
Lotteries over $o_1 \succ o_2 \succ o_3$	$x = (p_1,\ p_1 + p_2)$ e.g. $(.5, .2, .3) \mapsto (.5, .7)$	mix the probabilities	o_1 with prob. t, else o_3

Preferences: Basic axioms

Preferences $\succsim$ on the mixture space $\mathcal{X}$:

- **Complete:** $x \succsim y$ or $y \succsim x$, for all x, y
- **Transitive:** $x \succsim y$ and $y \succsim z \Rightarrow x \succsim z$
- **Mixture continuous:** $\{\alpha : x\alpha y \succsim z\}$ and $\{\alpha : z \succsim x\alpha y\}$ are closed
- **Strictly monotone:** $x \geq y \Rightarrow x \succsim y$, and $x > y \Rightarrow x \succ y$
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Irrational, incomplete, or stochastic choice can imply strange mixture attitudes.
Our focus: intentional, rational choice.

Basic mixture attitudes

How have we thought about mixing so far? **One-sided.**

Definition (Convex: mixture loving)

For all $x \succsim y$ and $\alpha \in [0,1]$,

$$x \alpha y \succsim y.$$

Definition (Concave: mixture averse)

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This is strong! We want a hierarchy that subsumes it and expands it.

A hierarchy of mixture attitudes: MMA

Definition (Mixed mixture attitudes, MMA)

For some $x^+ \succsim y^+$, $x^- \precsim y^-$ and $\alpha, \beta \in (0,1)$,

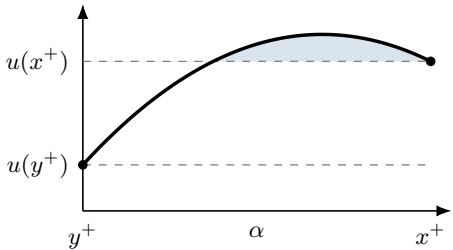
$$x^+ \alpha y^+ \succ x^+, \qquad x^- \beta y^- \prec y^-.$$

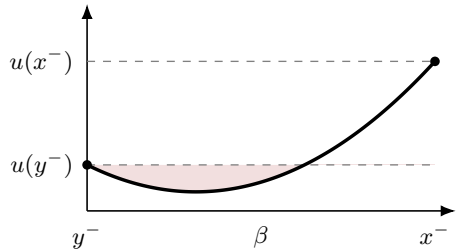
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A hierarchy of mixture attitudes: WNB

Definition (Weak non-betweenness, WNB)

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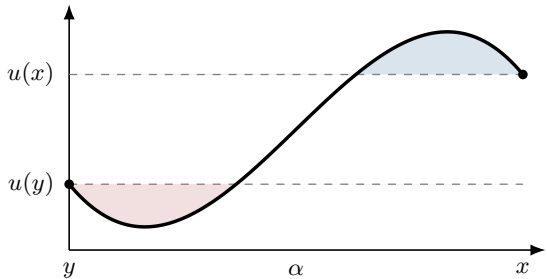
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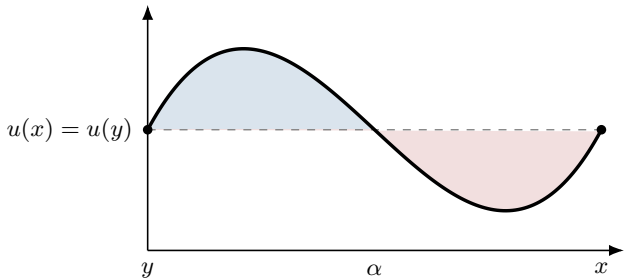
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How big is each set?

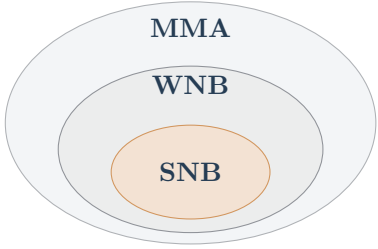
Taking stock of the hierarchy

MMA $x^+\alpha y^+ \succ x^+$ and $x^-\beta y^- \prec y^-$,

for some $x^+ \succsim y^+$, $x^- \succsim y^-$

WNB $x\alpha y \succ x \succsim y \succ x\beta y$, for some $x \succsim y$

SNB $x\alpha y \succ x \sim y \succ x\beta y$, for some $x \sim y$



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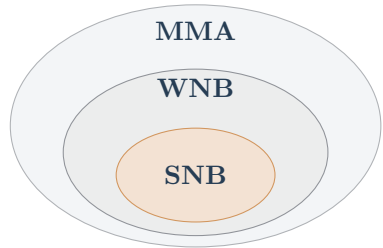
Taking stock of the hierarchy

MMA $x^+ \alpha y^+ \succ x^+$ and $x^- \beta y^- \prec y^-$,

for some $x^+ \succsim y^+$, $x^- \succsim y^-$

WNB $x\alpha y \succ x \succsim y \succ x\beta y$, for some $x \succsim y$

SNB $x\alpha y \succ x \sim y \succ x\beta y$, for some $x \sim y$



One-sided attitudes: "$\succsim$" or "$\sim$" in a definition makes no difference.
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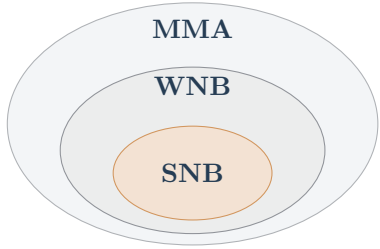
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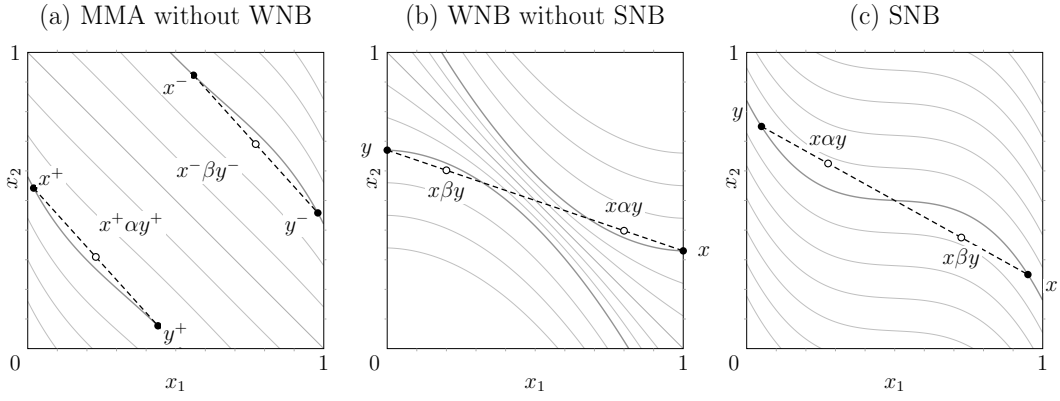
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What is in MMA without WNB? In WNB without SNB? Are these sets big?

Three example indifference maps

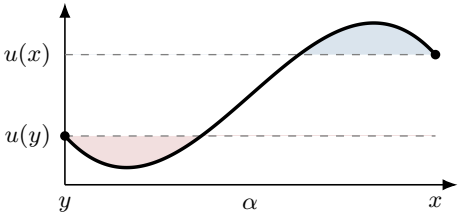


MMA $\supsetneq$ WNB $\supsetneq$ SNB.

WNB vs. SNB: the order of peak and dip

Proposition (Ordered attitudes, informal)

Without SNB, a chord $x \succ y$ with both attitudes must dip near y before it peaks near x.

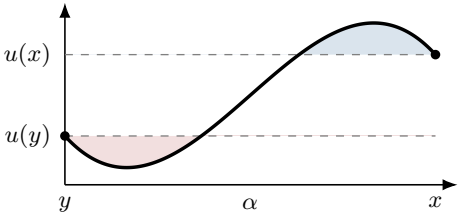


Dip, then peak: WNB without SNB

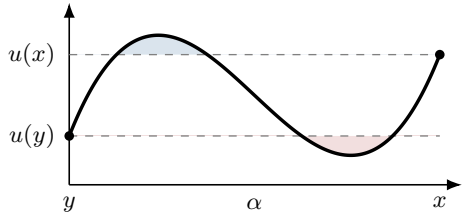
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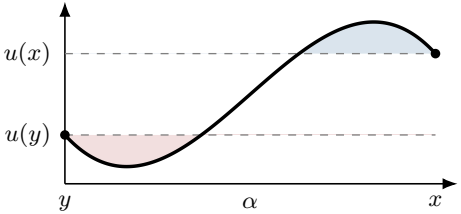


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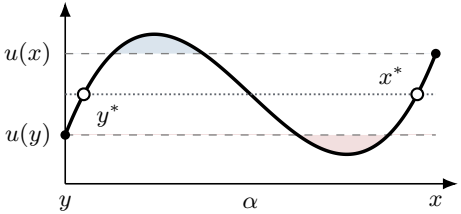
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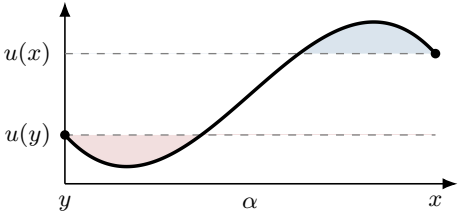


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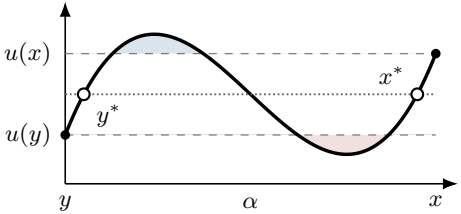
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Dip, then peak: WNB without SNB



Peak, then dip: always SNB

Mixed attitudes without SNB are restrictive.
Is SNB pathological / knife-edge?

SNB is generic

- **Calibrated utility:** $U_{\succsim}(x) = t$ iff $x \sim (t, \ldots, t)$
- **Closeness** (C^k): close calibrated utilities and first k derivatives

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Theorem (Genericity)

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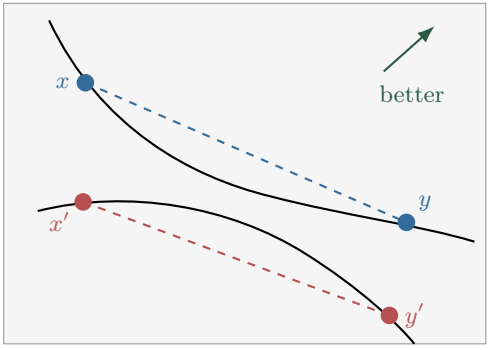
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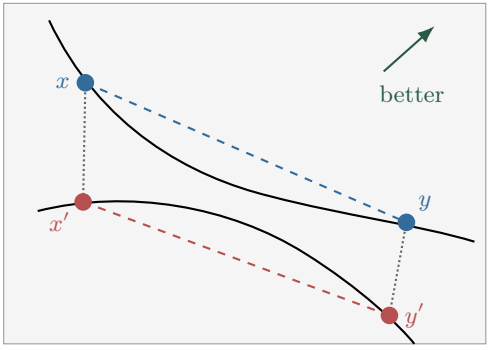
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2. *SNB preferences are* ***not dense*** *among preferences satisfying the basic axioms, for C^k preferences with $k \geq 2$.*

Why SNB is generic: proof sketch



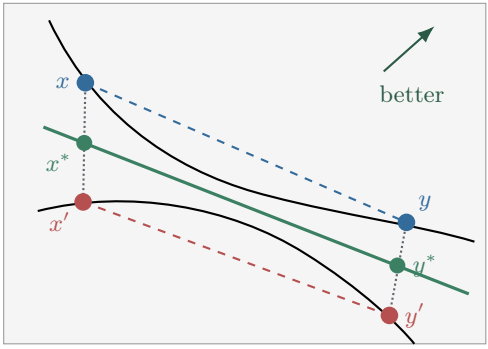
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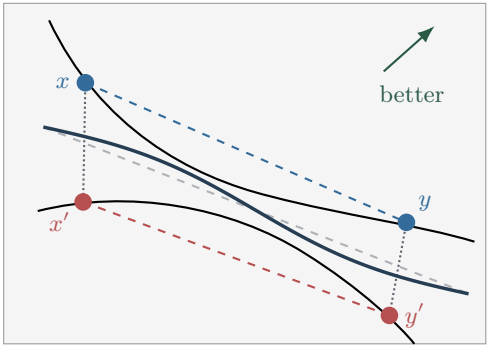
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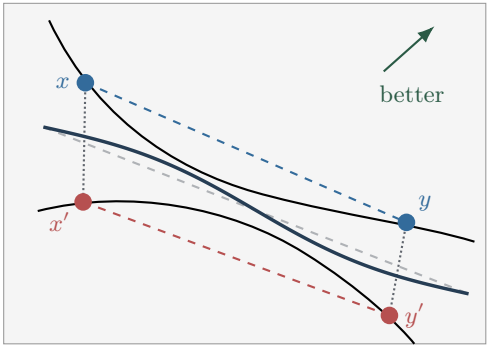
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SNB comes from distorting a neutral chord:

1. Hence the density result is robust, and special to mixed attitudes
2. What about models that impose neutrality in a structured way?

When neutrality is built in

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Certainty independence: every constant act is a neutral benchmark.
α-maxmin, Choquet and invariant biseparable: MMA $\iff$ SNB.

Classifying models

Uncertainty	MMA, not WNB	WNB, not SNB	SNB
α-maxmin EU	–	–	✓
Choquet EU	–	–	✓
Invariant biseparable	–	–	✓
Smooth ambiguity	✓	✓	✓
Variational, no UA	✓	✓	✓

Certainty-independent models display mixed attitudes only as SNB.

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Risk	MMA, not WNB	WNB, not SNB	SNB
Quadratic utility	✓	–	–
Mixture symmetry	✓	✓	–
RDEU, $n \geq 4$	–	–	✓
RDEU, $n = 3$	✓	–	✓
Cumulative prospect theory	✓	✓	✓

- Masatlioglu & Raymond (2016): Quadratic utility $\cap$ RDEU$\neq \emptyset$
- We show: only if mixture attitudes are globally one-sided

▸ Quadratic utility ▸ All models

Why does mixed mixture attitude arise?

- Theory is explicitly *general* and *descriptive* – *why* may be domain dependent
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- Experiment focuses on a specific environment to discuss particular channels
- General mechanism: rich mixture attitudes arise out of deliberate aggregation of multiple motives

Experiment

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3. For WNB and SNB: several mixtures on one chord

Most designs observe a single interior mixture

Audit of experimental literature

Domain	*N*	**Convex/ concave**	**Strict loving or averse only**	**MMA**	**WNB/SNB**
Randomization and lotteries	54	**25** (46%)	**15** (28%)	**4** (7%)	**4** (7%)
Consumption budgets	21	**1** (5%)	**8** (38%)	**1** (5%)	**4/3** (19/14%)
Time preferences	83	**67** (81%)	**5** (6%)	**3** (4%)	0
Portfolio and asset choice	97	**73** (75%)	**13** (13%)	**1** (1%)	0
Charitable giving	106	**88** (83%)	**3** (3%)	0	**3** (3%)
Effort allocation	43	**21** (49%)	0	0	0
Social and interpersonal allocation	185	**142** (77%)	**18** (10%)	0	**8** (4%)
All domains	**589**	**417** (71%)	**62** (11%)	**9** (2%)	**19/18** (3/3%)

Audit of experimental literature: details

28 of 589 designs allow, in principle, a direct test of MMA / WNB / SNB

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8 of 28 studies have information to estimate a **lower bound** on prevalence:

Study	**Domain**	N	**MMA**	**WNB / SNB**
Hey & Strazzera (1989)	lotteries	9	89%	78%
McGranaghan et al. (2025)	lotteries	2,102	47%	–
MacCrimmon & Toda (1969)	consumption	7	14%	14%
Bruyneel et al. (2017)	consumption	50	4%	–
Murphy et al. (2011)	social	100	18%	15%
Crawford & Harris (2018)	social	160	0.6%	0.6%
Epper et al. (2024) / Fehr et al. (2023)	social	815	10%	8%
Fehr et al. (2026)	social	3,352	52%	46%

Experimental environment

We revisit Toussaert (2024)

Subjects choose between trip destinations and lotteries over trip destinations

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Empirical relevance: surprise-trip planning was a USD 1.4 billion market in 2024

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Parts 3 and 4 of the experiment

Structure of the study

83 participants, London, March 2020. Online survey, 44 minutes on average

Part 1 **Trip preferences**

- Rank the 10 trips
- Value each trip, £0–500

Part 1: ten trips

Destination: **Turku** (Finland)





Description: Step back through the ages in this medieval town, and explore the maze-like chambers of the impressive Turku Castle. Enjoy nature in national parks, sunbathe on beaches, and explore museums, art galleries, historic sites, restaurants aplenty, and shopping hotspots, all around town.

Accommodation: Arrive at Helsinki airport and take a train to Turku. For the best experience of living local, stay in a cosy Airbnb that you'll have entirely to yourself, located conveniently in the central of town.

Ten European destinations

Düsseldorf	Prague
Gothenburg	Santiago
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All worth £420: cheaper travel means better accommodation.

▸ Valuations

Eliciting rankings

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- **Verify** with binary choices between sure trips

Strict ranking confirmed for 84% of adjacent pairs.

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Part 2 **45 binary choices**

- 6: sure trip vs. sure trip
- 29: sure trip vs. lottery, $p \in \{.1, .5, .9\}$
- 10: larger lotteries, held out

Choices: sure trips vs. surprise lotteries

After ranking the trips, $x_1 \succ x_2 \succ \cdots \succ x_{10}$, face binary choices, including:

- x_1 vs. $p\,x_1 + (1-p)\,y$ and y vs. $p\,x_1 + (1-p)\,y$
- $p \in \{0.1,\, 0.5,\, 0.9\}$
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- Surprise: the destination is revealed in the week of departure

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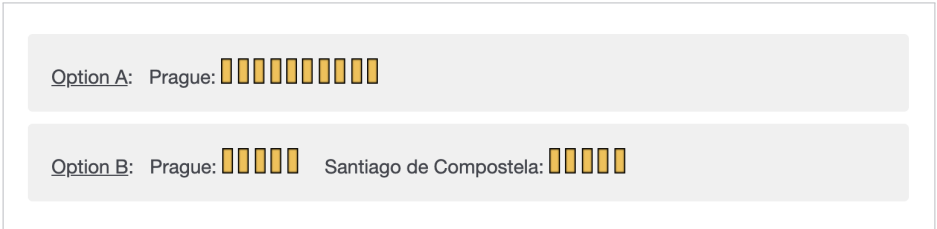
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Each mixture is observed at three weights, against both endpoints.

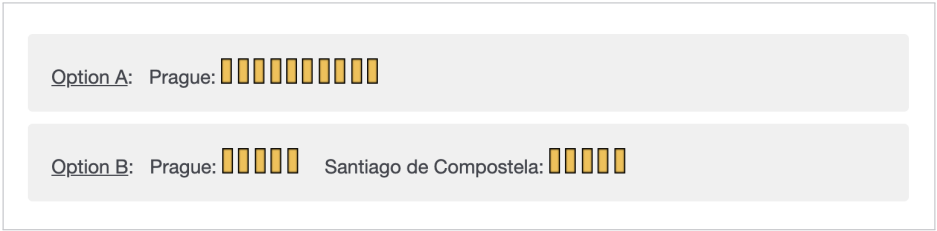
▸ All 45 problems

Part 2: what a choice looks like



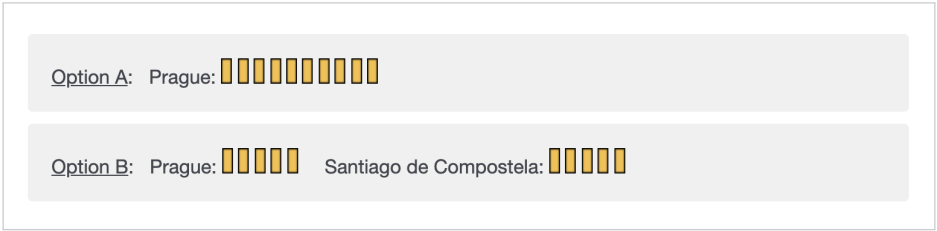
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- Suggested: open it at the airport

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Part 3

later

Part 4

later

One decision is paid: 1 in 20 wins a trip. Everyone receives a £25 voucher.

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Definition (SNB*)

For some $x \gtrsim y$ and $\alpha < \beta$ in $(0,1)$,

$$x\alpha y \succ x \succ x\beta y \qquad \text{or} \qquad x\alpha y \succ y \succ x\beta y.$$

Testing SNB

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Hard to test: it needs an exact indifference $x \sim y$, which choices rarely reveal.

Definition (SNB*)

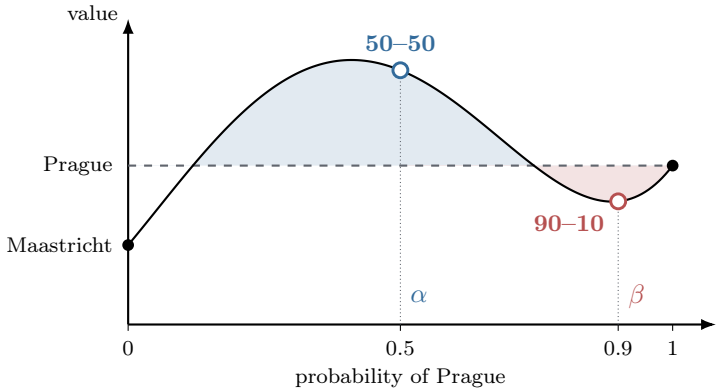
For some $x \succsim y$ and $\alpha < \beta$ in $(0,1)$,

$$x\alpha y \succ x \succ x\beta y \qquad \text{or} \qquad x\alpha y \succ y \succ x\beta y.$$

Proposition

Under the basic axioms, SNB $\iff$ *SNB**.

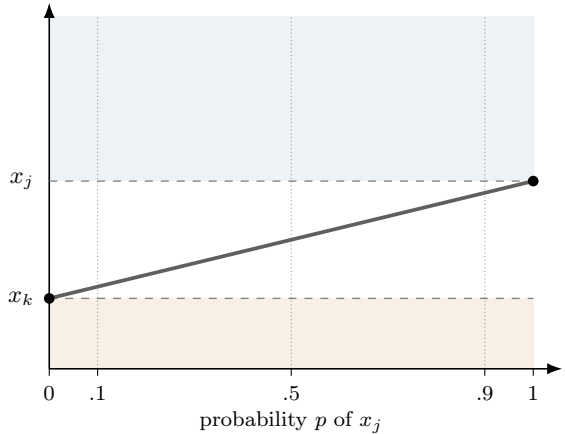
Graphical illustration



$$\alpha < \beta \;\Rightarrow\; \text{SNB}^* \iff \text{SNB}$$

From choices to types

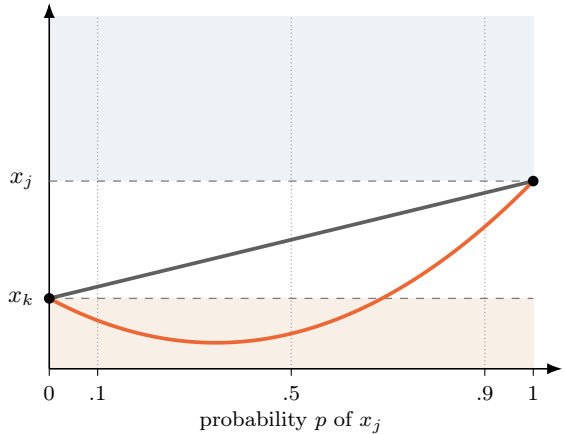
On a chord $x_j \succ x_k$, with $L(p) = (x_j, p;\, x_k, 1-p)$:



Betweenness
no lottery outside $[x_k, x_j]$

From choices to types

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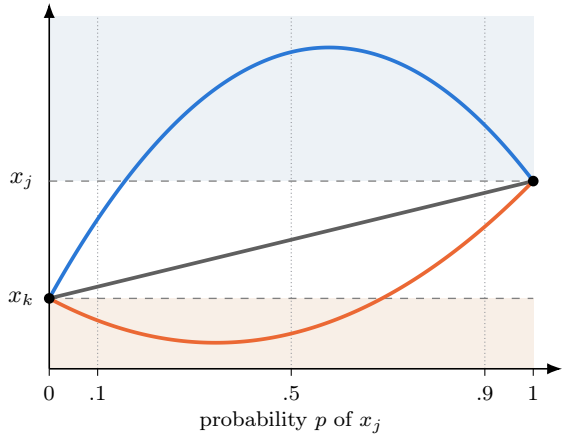


Betweenness
no lottery outside $[x_k, x_j]$

Averse only
$x_k \succ L(p)$ for some p

From choices to types

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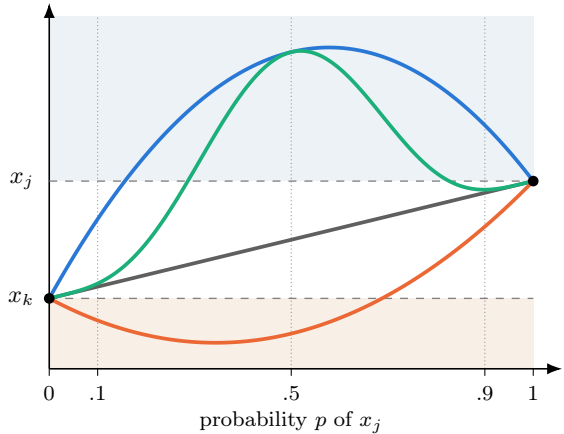
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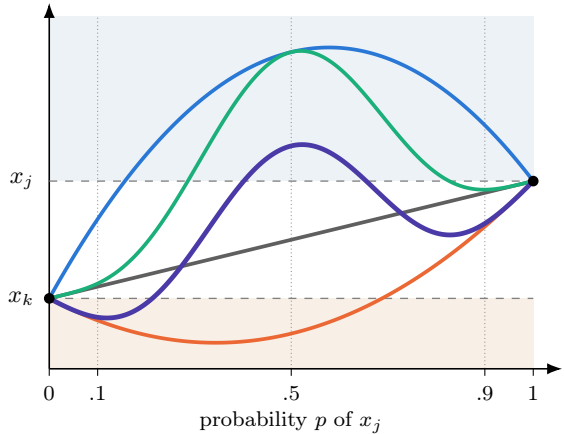
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Loving only
$L(p) \succ x_j$ for some p

SNB, no dip
$L(p) \succ x_j \succ L(p')$, $p < p'$
e.g. $L(.5) \succ x_j \succ L(.9)$

From choices to types

On a chord $x_j \succ x_k$, with $L(p) = (x_j, p;\, x_k, 1-p)$:



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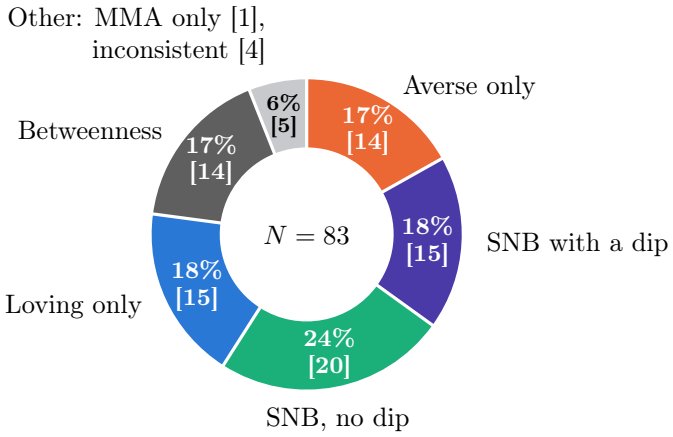
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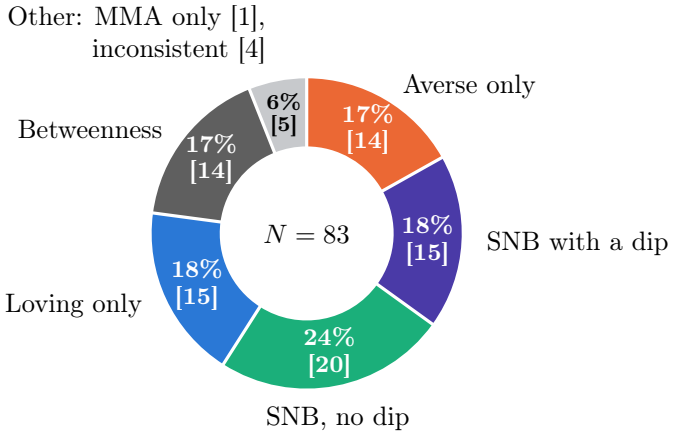
SNB with a dip
SNB, and also $x_k \succ L(p)$

RQ1: How rich are mixture attitudes?

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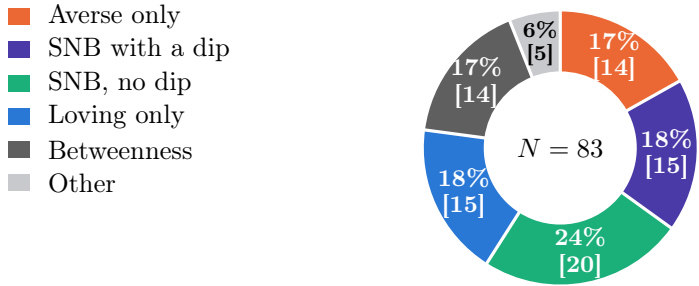
RQ1: How rich are mixture attitudes?



42% display SNB. The rest split evenly: betweenness, averse only, loving only.

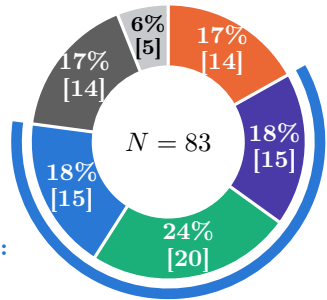
▸ Strictness

RQ1: How rich are mixture attitudes?



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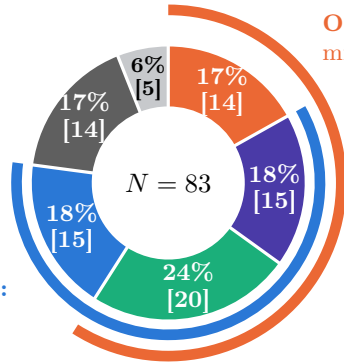
- Averse only
- SNB with a dip
- SNB, no dip
- Loving only
- Betweenness
- Other



Only 50–50 observed:
misidentified as loving

RQ1: How rich are mixture attitudes?

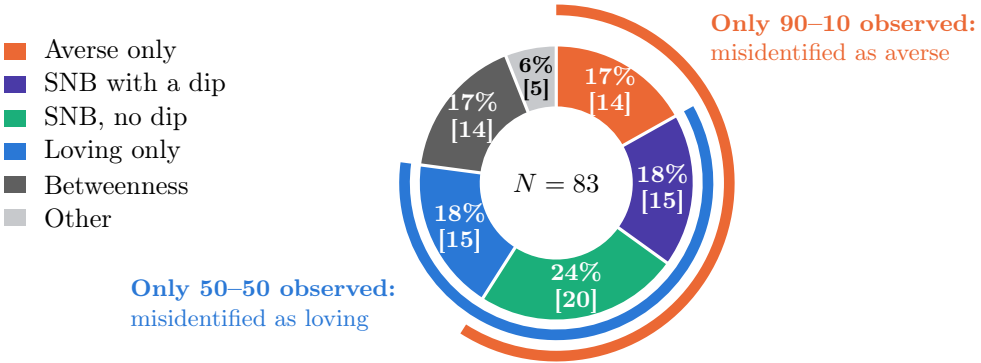
- Averse only
- SNB with a dip
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Only 90–10 observed:
misidentified as averse

Only 50–50 observed:
misidentified as loving

RQ1: How rich are mixture attitudes?



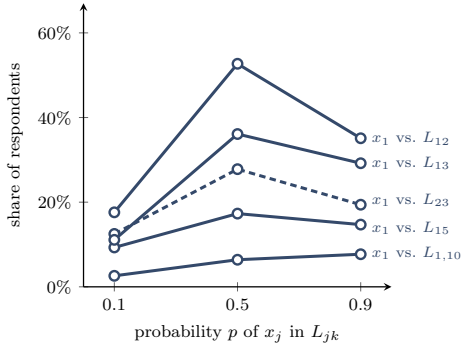
Observing a single mixture misreads SNB as a one-sided attitude.

RQ2: Which mixtures do people prefer?

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Share of participants; $L_{jk} = (x_j, p;\, x_k, 1-p)$

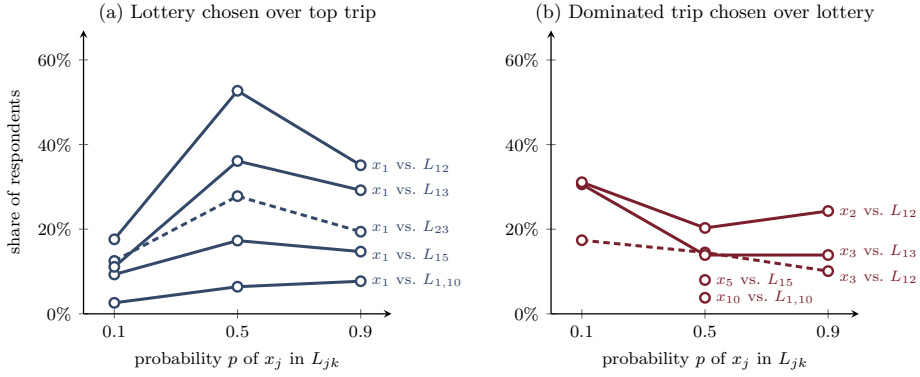
(a) Lottery chosen over top trip



▸ A mistake? ▸ Regressions

RQ2: Which mixtures do people prefer?

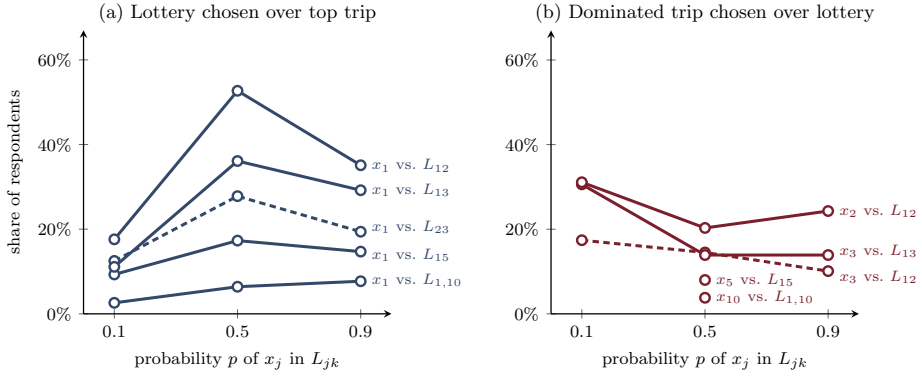
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▸ A mistake? ▸ Regressions

RQ2: Which mixtures do people prefer?

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Mixing helps most at 50–50, and hurts most at 10–90.

▸ A mistake? ▸ Regressions

Structure of the study

83 participants, London, March 2020. Online survey, 44 minutes on average

Part 1 **Trip preferences**

- Rank the 10 trips
- Value each trip, £0–500

done

Part 2 **45 binary choices**

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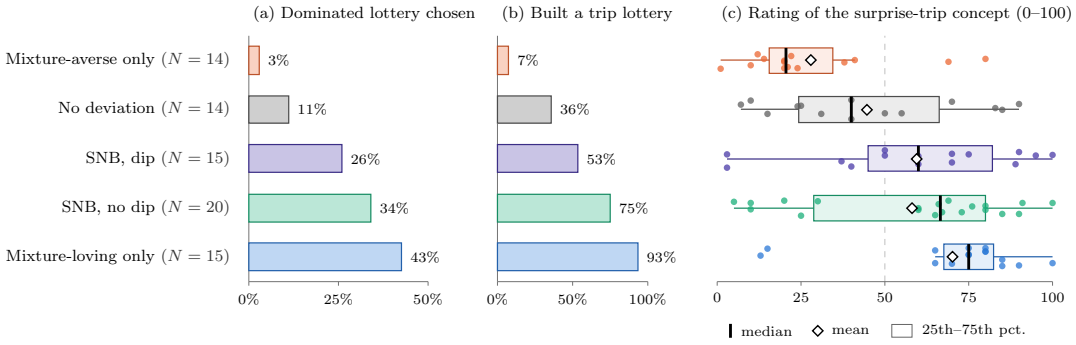
Part 4

later

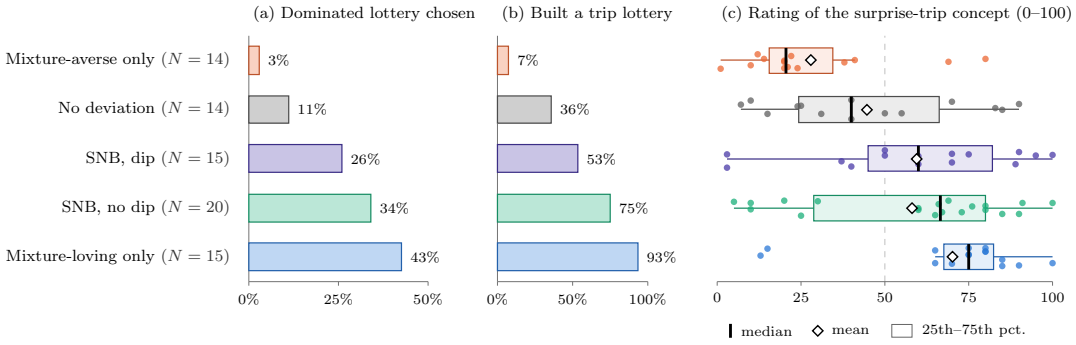
One decision is paid: 1 in 20 wins a trip. Everyone receives a £25 voucher.

RQ3: Is mixing behavior intentional or noisy?

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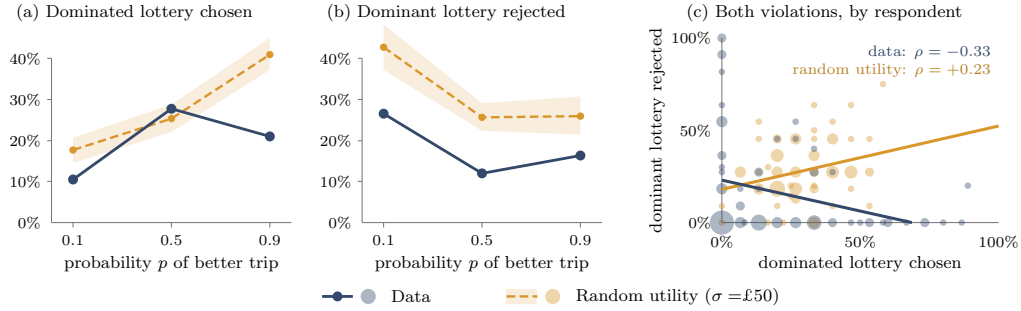
RQ3: Is mixing behavior intentional or noisy?



Types predict behavior in other choices.

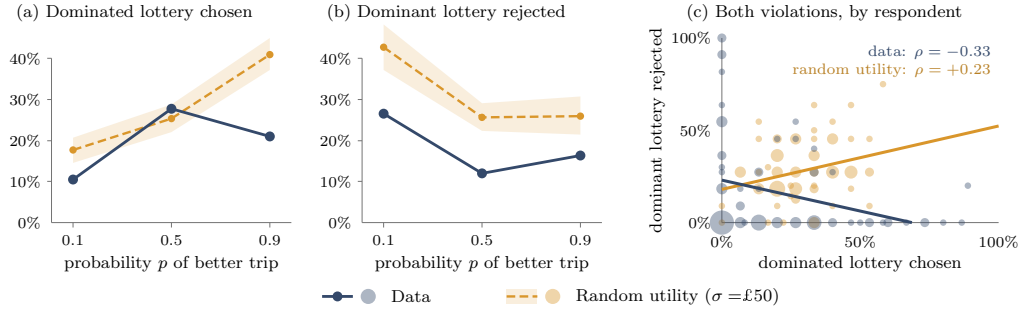
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▸ Benchmarks

RQ3: Is mixing behavior intentional or noisy?



Behavior cannot be rationalized by noise.

▸ Benchmarks

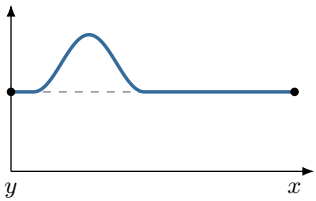
Rich motives generate SNB

Lotteries between x and y: p = probability of x
Two one-sided motives, loving $L(p)$ and averse $A(p)$; $U(p) = F\big(L(p), A(p)\big)$,
with F continuous, strictly increasing and $F(t,t) = t$

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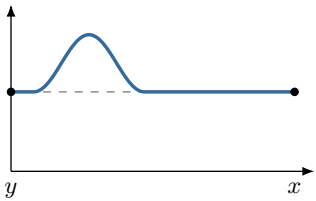
Loving motive L



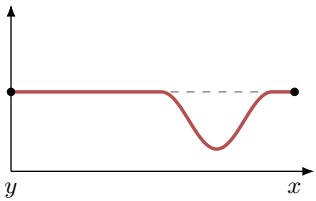
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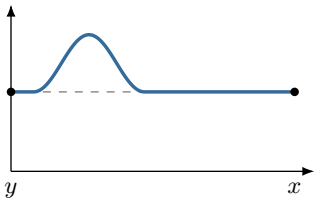
Averse motive A



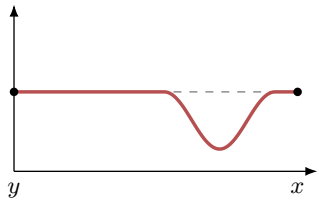
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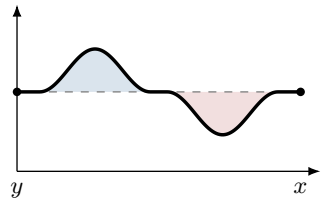
Loving motive L



Averse motive A



Aggregate U

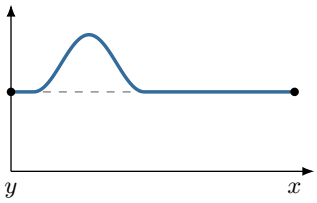


peak, then dip: SNB

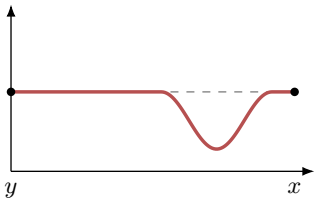
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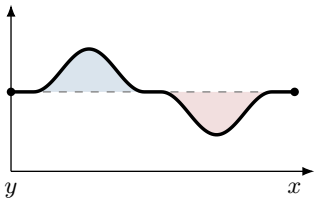
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Aggregate U



peak, then dip: SNB

Proposition

There exist motives L and A such that U displays SNB for every such F.

What preferences can aggregation of opposite motives generate?

$x \succ y$, $p =$ probability of x.

Proposition (Informal statement)

The following are **equivalent**:

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The following are **equivalent**:

1. **Utility:** $U(p) = V(p) + H(p)$, with $H(p) = H(1-p)$ and $H(0) = H(1) = 0$

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1. **Utility:** $U(p) = V(p) + H(p)$, with $H(p) = H(1-p)$ and $H(0) = H(1) = 0$
2. **Motives:** $U(p) = F\big(V(p) + S(p),\, V(p) - C(p)\big)$, where
   - S, C: continuous, symmetric about $\frac{1}{2}$, zero at both endpoints
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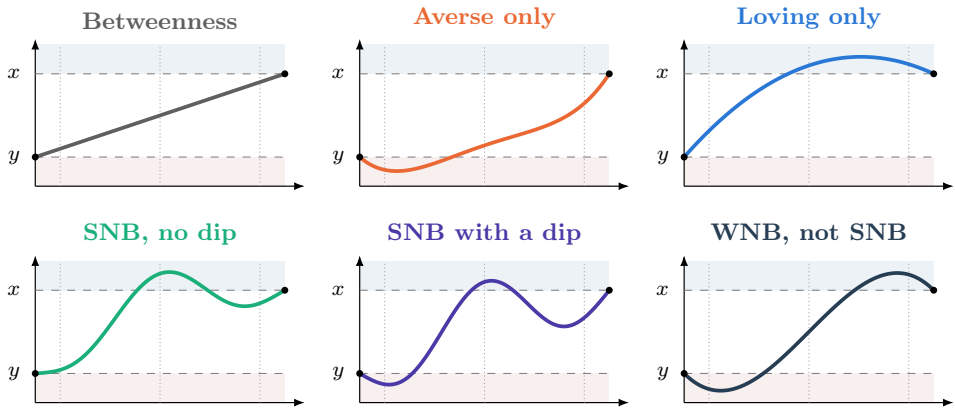
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3. **Patterns:** only the patterns observed in the experiment are possible

Patterns compatible with $U(p) \geq U(1-p)$ in the experiment



RQ4: Where does the (dis)taste for randomization come from?

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Share of participants ranking each reason first

What people like

rate the concept 50 or more ($N = 48$)

Excitement until the reveal	**33%**
The unique holiday experience	**27%**
Daydreaming about destinations	**19%**

▸ Correlation: motives, choices and wildcard valuation ▸ Lottery delay results

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Two forces: open questions are fun, but hard to prepare for.

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Part 4 **Money and tastes**

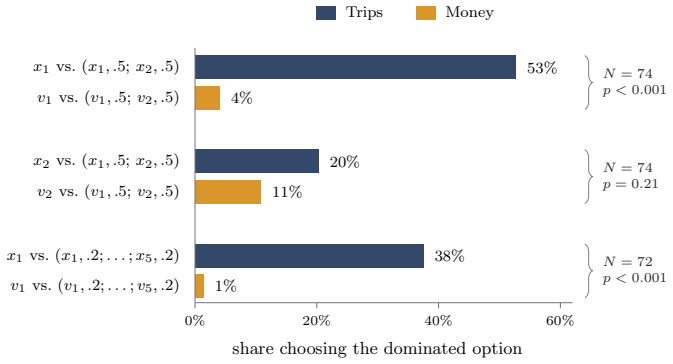
- Same choices over money
- Risk, ambiguity, trip attributes

One decision is paid: 1 in 20 wins a trip. Everyone receives a £25 voucher.

RQ5: How does multidimensionality affect mixing?

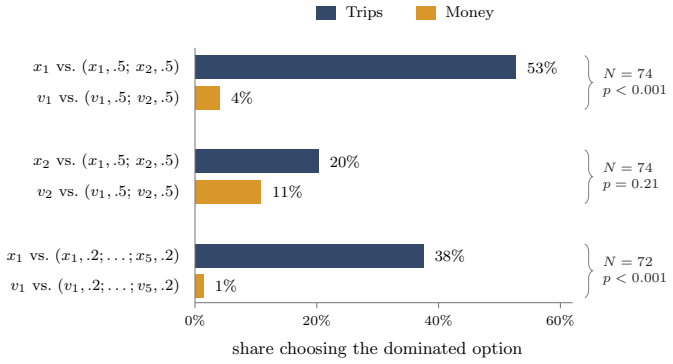
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Same choices, with each trip x_k replaced by its own valuation v_k



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Money is one-dimensional: mixing disappears, 53% → 4%.

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Why? Trips are bundles of attributes

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  - **Vertical**: a clear preference for one pole
  - **Horizontal**: open to either experience
- Vertical attributes rank only 9% of trip pairs;
  90% of subjects have a horizontal attribute

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A simple model: what a lottery L over two trips is worth

$$V(L) = \underbrace{\mathbb{E}_L[v]}_{\text{value}} + \underbrace{S(h)}_{\text{fun of not knowing}} - \underbrace{C(h)}_{\text{cost of not preparing}}$$

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- v: **value** of a trip, from its vertical attributes

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- v: **value** of a trip, from its vertical attributes
- h: **uncertainty** about horizontal attributes:
  mixing over horizontal attributes creates mixture attitudes

RQ5: How does multidimensionality affect mixing?

A simple model: what a lottery L over two trips is worth

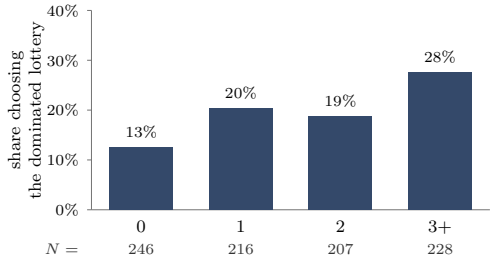
$$V(L) = \underbrace{\mathbb{E}_L[v]}_{\text{value}} + \underbrace{S(h)}_{\text{fun of not knowing}} - \underbrace{C(h)}_{\text{cost of not preparing}}$$

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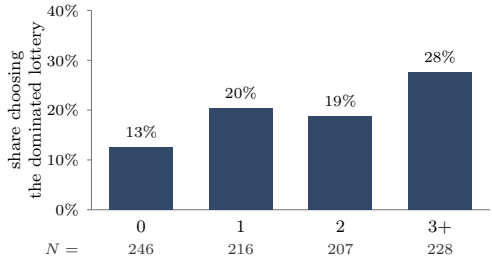
Horizontal attributes: mixing



horizontal attributes on which the two trips differ

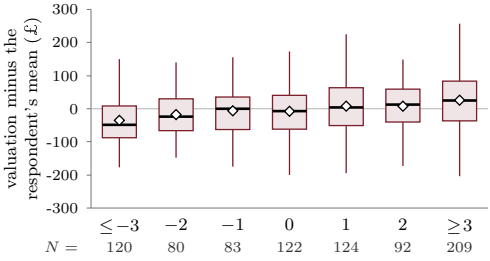
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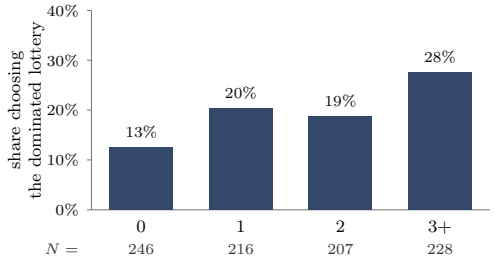
Vertical attributes: value



taste fit of the trip on vertical attributes

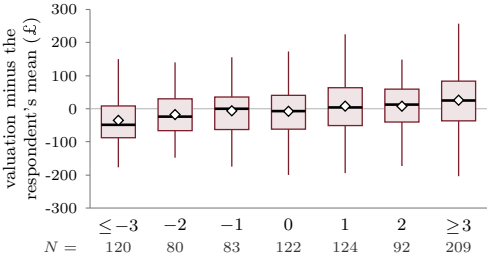
RQ5: How does multidimensionality affect mixing?

Horizontal attributes: mixing



horizontal attributes on which the two trips differ

Vertical attributes: value



taste fit of the trip on vertical attributes

Horizontal differences drive mixing: +3.8 points each.
Vertical fit drives value: +£8.4 per point.

Conclusion

Mixture attitudes are likely richer than we thought:

1. **Theoretically**, rich taxonomy, not knife-edge
2. **Experimentally**, common but understudied

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Mixture attitudes are likely richer than we thought:

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Mixture attitudes capture the discrepancy between preference and mixture operation.

- Richness of attitude captures deliberate choice + richness in environment
- Contrast with complex low-stakes choices $\Rightarrow$ global mixture preferences
- A taste for randomization when randomness is high, traded off against a preference for certainty near 100–0

Further work

We would love your feedback

Further work

We would love your feedback

Experiments

What kind of additional experiments would you like to see?

Further work

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Theory

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Domains

In which mixture domains would it be nice to run this experiment?
consumption, social allocations, time, information, strategies, ...

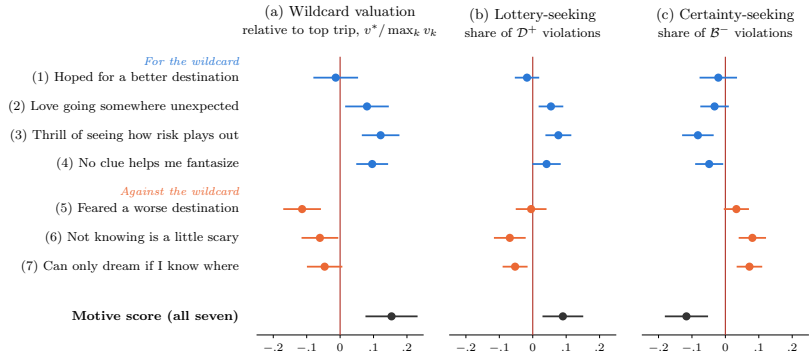
Thank You!

Appendix

RQ4: Where does the (dis)taste for randomization come from?

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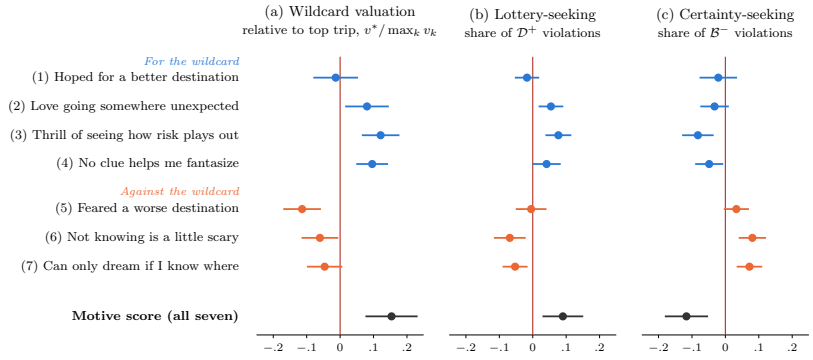
Value a **wildcard** trip to an unknown destination; rate motives for and against



Back

RQ4: Where does the (dis)taste for randomization come from?

Value a **wildcard** trip to an unknown destination; rate motives for and against



Motives for/against uncertainty predict wildcard value and lottery choice.

Back

RQ4: Where does the (dis)taste for randomization come from?

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Part 3: build your own lottery, then choose when to learn the destination

46 of 83

build a lottery

Lottery building Back

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74%

of builders delay the reveal, by 94 days on average

Lottery building Back

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Lottery building Back

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Part 3: build your own lottery, then choose when to learn the destination

46 of 83

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74%

of builders delay the reveal, by 94 days on average

54%

of builders give up at least £5 to keep the surprise

People pay to keep the surprise, but their lotteries favor top trips and drop the worst.

Lottery building Back

Why quadratic utility stops at MMA

Quadratic utility: $V(\mathbf{p}) = \mathbf{p}' \, \Phi \, \mathbf{p}$. Along any chord,

$$V(\mathbf{p}\alpha\mathbf{q}) = \alpha V(\mathbf{p}) + (1-\alpha)V(\mathbf{q}) + K(\mathbf{p},\mathbf{q})\,\alpha(1-\alpha), \qquad K(\mathbf{p},\mathbf{q}) = -(\mathbf{p}-\mathbf{q})' \, \Phi \, (\mathbf{p}-\mathbf{q}).$$

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- The sign can change across chords: MMA, but never WNB

Attitudes depend on *which* lotteries are mixed, not on *how much*.
So 50–50 surprise $\succ$ Prague $\succ$ 90–10 surprise is ruled out.

Back

Which attitudes can each model capture?

Model	Key feature	MMA, not WNB	WNB, not SNB	SNB
Risk				
Quadratic utility (Machina 1982; Chew et al. 1991)	quadratic in probabilities	✓	–	–
Mixture symmetry (Chew et al. 1991)	symmetry on indifferent chords	✓	✓	–
RDEU, $n = 3$ (Quiggin 1982)	common probability weighting	✓	–	✓
RDEU, $n \geq 4$ (Quiggin 1982)	common probability weighting	–	–	✓
RDEU, Prelec w, $n \geq 3$ (Prelec 1998)	parametric probability weighting	–	–	✓
Bell disappointment, $n \geq 3$ (Bell 1985)	linear, mean-referenced disappointment	–	–	✓
Pure risk (Dillenberger et al. 2026)	domain-specific risk attitudes	✓	–	✓
CPT, unrestricted weights (Tversky & Kahneman 1992)	separate gain/loss weighting	✓	✓	✓
Uncertainty				
α-maxmin EU (Hurwicz 1951)	weighted best/worst expected utilities	–	–	✓
Choquet EU (Schmeidler 1989)	nonadditive beliefs	–	–	✓
Invariant biseparable (Ghirardato et al. 2004)	certainty independence	–	–	✓
Smooth ambiguity (Klibanoff et al. 2005)	unrestricted increasing aggregator	✓	✓	✓
Variational minus UA (Chandrasekher et al. 2022)	optimism over penalized prior sets	✓	✓	✓
Vector EU (Siniscalchi 2009)	symmetric ambiguity adjustment	✓	✓	✓

A check: some specification belongs to the set; a dash: none does. n: number of ordered prizes. Uncertainty rows: bounded statewise utility-profile domains. UA: uncertainty aversion.

Back

A mistake?

Read alone, 50–50 $\succ$ Prague violates first-order stochastic dominance in the participant's own ranking (**?**). But these choices are:

Costly	$\approx$**10%** of a trip's value given up to keep the surprise
Systematic	**53%** choose the surprise at 50–50, but only **35%** at 90–10
Valued	**54%** of lottery builders pay $\geq$ £5 to delay the reveal
Specific to trips	**4%** choose the 50–50 lottery when prizes are money

Trips are bundles. All cost £420, so a cheaper city means a better hotel. Without attribute-wise dominance, monotonicity is silent. Money is the one-attribute limit.

Back

A choice from our experiment

Sure trip

your favorite destination

(say, Prague)

vs.

50–50 surprise
Prague or your 2nd favorite
(say, Maastricht)

53%
choose the surprise

90–10 surprise
90% Prague, 10% Maastricht

35%
choose the surprise

Many make both choices: 50–50 surprise $\succ$ Prague $\succ$ 90–10 surprise

How strict is the SNB classification?

Choices reveal weak preferences. We read choices at the design points as strict.

SNB participants	**35**
Strict by transitivity through strictly ranked trips	9
Overturned only by two exact ties	8
Overturned by one exact tie	18

Most conservative reading: 9 of 83 participants (11%) display SNB.

Back

The 45 decision problems

DP #	Core problems (35 choices)	**Use**
	Degenerate lotteries (6 choices)	
1, 4, 10, 13	A: $(x_1, 1)$ vs. B: $(x_k, 1)$ for $k \in \{2, 3, 5, 10\}$	Ranking
7	A: $(x_2, 1)$ vs. B: $(x_3, 1)$	Ranking
16	A: $(x_5, 1)$ vs. B: $(x_{10}, 1)$	Ranking
	Binary $\mathcal{D}^+$ problems (15 choices)	
2, 18, 27	A: $(x_1, 1)$ vs. B: $(x_1, p; x_2, 1-p)$ for $p \in \{0.5, 0.9, 0.1\}$	ML
5, 20, 29	A: $(x_1, 1)$ vs. B: $(x_1, p; x_3, 1-p)$ for $p \in \{0.5, 0.9, 0.1\}$	ML
8, 22, 31	A: $(x_1, 1)$ vs. B: $(x_2, p; x_3, 1-p)$ for $p \in \{0.5, 0.9, 0.1\}$	ML*
11, 24, 33	A: $(x_1, 1)$ vs. B: $(x_1, p; x_5, 1-p)$ for $p \in \{0.5, 0.9, 0.1\}$	ML
14, 25, 34	A: $(x_1, 1)$ vs. B: $(x_1, p; x_{10}, 1-p)$ for $p \in \{0.5, 0.9, 0.1\}$	ML
	$\mathcal{B}^-$ problems (11 choices)	
3, 19, 28	A: $(x_2, 1)$ vs. B: $(x_1, p; x_2, 1-p)$ for $p \in \{0.5, 0.9, 0.1\}$	MA
6, 21, 30	A: $(x_3, 1)$ vs. B: $(x_1, p; x_3, 1-p)$ for $p \in \{0.5, 0.9, 0.1\}$	MA
9, 23, 32	A: $(x_3, 1)$ vs. B: $(x_1, p; x_2, 1-p)$ for $p \in \{0.5, 0.9, 0.1\}$	MA*
12	A: $(x_5, 1)$ vs. B: $(x_1, p; x_5, 1-p)$ for $p = 0.5$	MA
15	A: $(x_{10}, 1)$ vs. B: $(x_1, p; x_{10}, 1-p)$ for $p = 0.5$	MA
	Sure trip between lottery outcomes (3 choices)	
17, 26, 35	A: $(x_5, 1)$ vs. B: $(x_1, p; x_{10}, 1-p)$ for $p \in \{0.5, 0.9, 0.1\}$	Level
	Out-of-sample problems (10 choices)	
	$\mathcal{D}^+$ over 3, 5 or 10 trips (9 choices)	
36	A: $(x_1, 1)$ vs. B: $(x_1, 0.4; x_2, 0.3; x_3, 0.3)$	ML
37	A: $(x_1, 1)$ vs. B: $(x_1, 0.4; x_3, 0.3; x_5, 0.3)$	ML
38	A: $(x_1, 1)$ vs. B: $(x_2, 0.4; x_3, 0.3; x_4, 0.3)$	ML*
39	A: $(x_5, 1)$ vs. B: $(x_6, 0.4; x_7, 0.3; x_8, 0.3)$	ML*
40	A: $(x_1, 1)$ vs. B: $(x_1, 0.2; x_2, 0.2; x_3, 0.2; x_4, 0.2; x_5, 0.2)$	ML
41	A: $(x_1, 1)$ vs. B: $(x_1, 0.2; x_3, 0.2; x_5, 0.2; x_7, 0.2; x_{10}, 0.2)$	ML
42	A: $(x_1, 1)$ vs. B: $(x_6, 0.2; x_7, 0.2; x_8, 0.2; x_9, 0.2; x_{10}, 0.2)$	ML*
43	A: $(x_5, 1)$ vs. B: $(x_6, 0.2; x_7, 0.2; x_8, 0.2; x_9, 0.2; x_{10}, 0.2)$	ML*
44	A: $(x_1, 1)$ vs. B: $(x_1, 0.1; x_2, 0.1; ...; x_9, 0.1; x_{10}, 0.1)$	ML
	Sure trip between lottery outcomes (1 choice)	
45	A: $(x_5, 1)$ vs. B: $(x_1, 0.1; x_2, 0.1; ...; x_9, 0.1; x_{10}, 0.1)$	–

Back

Mixture attitudes by probability

	Lottery chosen over x_j (upper)			x_k chosen over lottery (lower)		
	(1)	(2)	(3)	(4)	(5)	(6)
$p = .9$	-6.7^{**}	-5.4^{*}	-5.5^{*}	-0.0	0.6	2.2
	(2.9)	(3.3)	(3.1)	(3.1)	(3.5)	(3.3)
$p = .1$	-17.3^{***}	-18.6^{***}	-18.5^{***}	10.2^{***}	9.6^{**}	8.1^{**}
	(3.1)	(3.4)	(3.6)	(3.6)	(3.9)	(4.1)
Value forgone (£100)		3.3	3.2		-3.3	-11.4
		(3.9)	(2.4)		(7.8)	(7.6)
Problem-triple FE	Yes	Yes	Yes	Yes	Yes	Yes
Respondent FE	No	No	Yes	No	No	Yes
Share at $p = .5$ (%)	27.8	27.8	27.8	12.0	12.0	12.0
Observations	1,113	1,113	1,113	798	798	798
Respondents	79	79	79	79	79	79

Linear probability models, percentage points; reference: $p = .5$. Standard errors clustered by respondent.

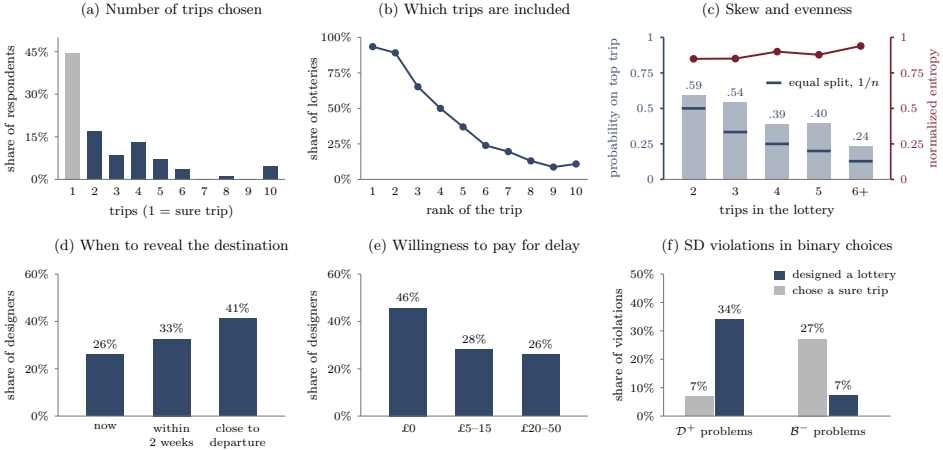
Back

Noise benchmarks

	Data	10% tremble	Random utility (σ =£50)	Person-specific tremble
Share of respondents by type (%)				
SNB, with a dip	18	51	78	67
		[43, 59]	[72, 84]	[60, 72]
SNB, without a dip	24	19	3	10
		[12, 27]	[0, 6]	[5, 16]
Mixture-loving only	18	7	3	3
		[2, 12]	[0, 6]	[0, 7]
Mixture-averse only	17	5	1	2
		[1, 8]	[0, 2]	[0, 5]
No deviation	17	9	1	11
		[4, 13]	[0, 2]	[7, 14]
Other[a]	6	9	14	8
		[6, 13]	[8, 18]	[5, 11]
Tests				
Type distribution, χ^2		66.6	643.5	196.0
		[1.0, 9.9]	[1.0, 9.4]	[0.6, 8.4]
Correlation of violations, ρ	−0.33	+0.01	+0.23	+0.60
		[-0.17, +0.21]	[+0.05, +0.41]	[+0.48, +0.71]
Out-of-sample gap[b] (pp)	+27.2	+0.1	+5.5	+20.0
		[-6.7, +6.1]	[-17.2, +20.5]	[+14.4, +24.9]

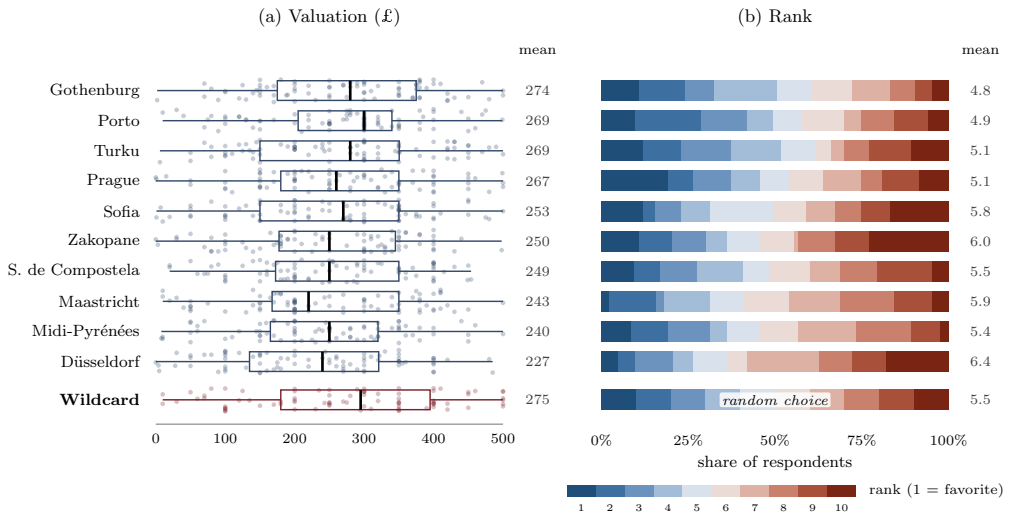
Back

Lottery building



Back

No objective winner



Back

Sample

Participants	83
Recruited through the LSE Behavioural Research Lab	87%
Male	59%
Age, mean (range)	27.1 (18–50)
Trips abroad in the last 12 months, mean	4.0
Of the 10 countries, visited before, mean	4.9
Days to the chosen travel date, mean	230
Completed after the Italian lockdown	51%
Survey time, mean	44 minutes

MMA without WNB: better bundles cost more

Proposition (Informal)

$\succsim$ displays no WNB if and only if, whenever prices $p \gg 0$ support the indifference sets of x and y,

$$x \succ y \quad \Longrightarrow \quad p \cdot x \;\geq\; p \cdot y.$$

Under one-sided attitudes, the better bundle costs more at supporting prices.
MMA without WNB keeps this. WNB is exactly where it fails.